EC441 Midterm Two Spring 2023 – SOLUTIONS

Q1. Subnets (25 points)

a. How many '/25' networks can be subnetted from a '/28' network?

None. The /28 is one-eighth the size of the /25

b. How many '/25' networks can be subnetted from a '/20' network?

$$2^{(25-20)} = 32$$

c. How many hosts can be configured in a '/64' IPv6 network?

Each `/64` network can handle 2^{64} hosts.

d. The following code prints out nothing. What are the values that the user typed for X and for Y?

```
import ipaddress as ip

X = input()
Y = input()

a = ip.ip_network("128.197.0.0/16")

b = [ip.IPv4Network('128.197.0.0/19'), ip.IPv4Network('128.197.32.0/19'),
  ip.IPv4Network(X+"/17"), ip.IPv4Network(Y+"/18")]


for net in b:
  if not net.subnet_of(a):
    print(net, "is not a subnet of", a)

for x in b:
  for y in b:
    if x != y and x.overlaps(y):
      print(x, y, "overlap")
```

The first "quadrant" of the `.0.0/16` has been used by the `/19` networks, up to `.0.63`. So, we need the next quadrant to be used by the `/18`, which would be `128.197.64.0/18`, and the rest (the second half of the space) would be used by the `/17` and it would be `128.197.128.0/17`

So

```
X = "128.197.128.0"   # /17
Y = "128.197.64.0"    # /18
```

Q3. TCP (25 points)

Consider a 40 Mb/s service provided by an ISP to customer Q. This Boston-based customer connects to a website using HTTP/TCP.

The packet size is 1500 bytes (ignore the 40 bytes of TCP/IP overhead). The round trip time for this server is 0.03 seconds.

The congestion window protocol being used is slow start (starts at one packet, increases by one every ACK)

a. How long will it take for the throughput of this connection not to be limited by the congestion control window?

At 40 Mb/s, it takes 0.3 milliseconds to send out one packet. So, in 30 ms, we could send 100 packets. We need the congestion window to be at least 100 packets.

It will grow like this:

```
RTT--V

     1    2    3     4     5     6     7

1 -> 2 -> 4 -> 8 -> 16 -> 32 -> 64 -> 128

CW-- ^
```

So, it will take 7 round trip times or 0.21 seconds for the congestion window not to be the limiting factor.

b. Does there need to be an adjustment to the receive window of Q in order to achieve a throughput of 40 Mb/s? If so, what adjustment?

The standard maximum receiver window is 65535 bytes, which only takes 13 ms to send. Hence, we need to adjust using *window scaling*, which makes each value of the receiver window advertised worth more by some factor or two. In linux, a scale of 512 is standard, and this would be more than sufficient to make this work.

c. The website says it is based in San Jose, which is 4000 km from Boston. Do you believe it?

The propagation time for 8000 km at 2×10^8 m/s is 40 ms. The website cannot be located in San Jose.

Q2. Error Control Coding (25 points)

a. Suppose the probability of bit error on a link is 10%. Design a link-layer protocol using block codes and an error control algorithm (i.e. some combination of error correction, detection, and retransmission) that will reduce the probability of error that the network layer sees to 1% or lower.

Your code should be as short as possible: i.e. for the (n, k) block code used, you should choose to minimize the value of n.

A `(2,1)` parity bit code with codewords 00 and 11 works and is the shortest code possible.

With this code, no errors occur 81% of the time.

One error occurs $(0.1)(0.9)(2) = 18$, and these are re-transmitted.

Two errors, which is not detectable, occurs 1% of the time.

When 00 or 11 arrives, this is translated back into 0 or 1. So, the net probablity of error using this code is exactly 1%.

b. Consider the following block code, as described by these codewords and bit map:

Data Bits	Transmitted Codeword
11	00000
10	01011
01	10101
00	11110

What is the rate of the code? What is the minimum distance of the code? If the receiver receives the code word 11111, what are its possible actions?

Your friend claims to have written an algorithm using this code that simultaneously corrects all single-bit errors and also asks for retransmissions whenever there are two bit errors. Do you believe them? Why or why not?

The rate of the code is 2/5. The minimum distance of the code is 3.

11111 is not a codeword. The receiver can

- discard it
- ask for a retransmission

- decode this as 00, since this is the closest codeword to the received word.

Since the minimum distance is 3 and we need $e_c + e_d < d_{min}$, we cannot implement error correction as well as 2-bit error detection.

Q4. Routing and Addressing (25 points)

a. Design a JSON / python / pseudocode format for (1) a link state (LS) packet and (2) a distance vector (DV) packet.

The link state packet needs this data:

```
LS = {"src":router, "id":idnumber, "links":[(addr1,cost1), (addr2,cost2)] }
```

```
DV = {"src":router, "costs":{addr:addr[cost] for addr in net} }}
```

Provide a network graph (make up your own connections and costs) for a network with six routers labelled A through F. Your network should have at least 8 full-duplex links.

Show an example of a LS packet and DV packet that node A would construct and send out using your packet formats defined above.

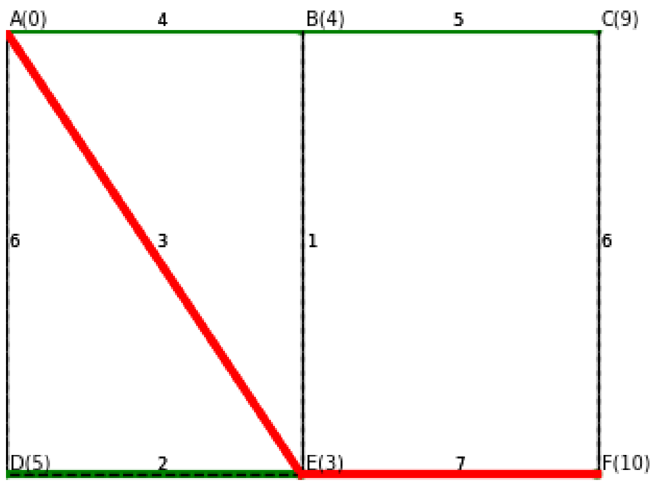

Figure 1: n1.png

```
LS = {"src":"A", "id":124, "links":[("B",4),("E",3),("D",6)]}
```

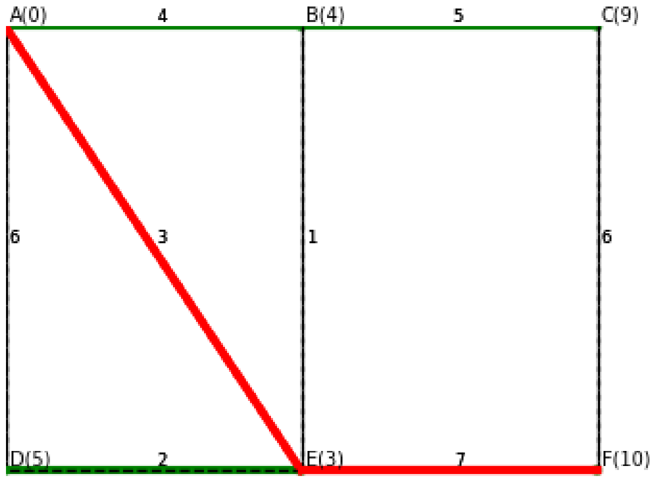

Figure 2: ns.png

```
DV = {"src":"A", "costs":{"A":0,"B":4,"C":9,"D":5,"E":3,"F":10} }
```

b. Explain the following networking terms by analogy to the world-wide system of postal mail.

Here is an example answer comparing "packet" to "letter"

——— 0 (example)

```
packet :: letter

A postal letter is equivalent to a network packet in that
- it has a source and destination address included in a prescribed format
- it is forwarded (inspected for destination)
over several "links" through its path.
- it sometimes gets delayed or discarded inside the network
```

——— 1

```
autonomous system :: national-postal service
```

Each national postal service, like USPS or CanadaPost, is responsible for routing of packets within its territory, and it can use its own mechanisms for doing this.

——— 2

```
BGP is how inter-AS routing works :: country-to-country routing
```

Each country announces port(s) of entry for its mail service, and other countries send international mail to one of those ports. It is also likely that mail for several countries (ASes) is bundled together and sent to regional hubs.

For example, mail to Europe is likely sent to, say, Germany first, then distributed from their to smaller countries.

——— 3

```
OSPF is an example of  how intra-AS routing works :: USPS routing
```

The USPS uses zip codes and street addresses to do the bulk of its routing. It uses larger regional distribution centers, then post offices, then letter carriers.

——— 4

```
hierarchical intra-AS routing :: Zip Codes
```

The zip code is the basic hierarchy of USPS addresses and the routing scheme. All mail for a certain zip code gets routed to the same post office (which is like a router). Internal mail within a zip code never leaves the post office: this is like level-2 switching within the subnet.

Zip codes themselves have a hierarchy. Massachussetts is mostly 01xxx and 02xxx.

——— 5

```
network address vs host address ::  the house vs the people inside
```

There are two possible analogies here. You could view the name on the envelope as the host address, and everything else is the network address. The final routing step is done by another layer : whoever collects the mail in the household distributes it to the occupants.

Another way of looking at this is the network address is the part of the address that specifies which letter carrier is given the letter. Then the host address is the details for that letter. The letter carriers use a different routing algorithm (walking their routes) than the post office does. This final step is like the link layer as it provides the last amount of routing to individual homes.